ABSTRACT DOMAINS AND CUT POINTS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: JOINS OF CLOSURE OPERATORS ON A LATTICE AND OF CUT-POINT TOPOLOGIES ON A CONTROL-FLOW GRAPH

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Abstract. An abstract domain, a closure operator ρ on a complete lattice, determines the class W_ρ of pairs $x \leq y$ that it collapses, an element of the lattice of two-out-of-three classes. We show that $W_\rho \cap W_{\rho'}$ is the class of the reduced product, that $W_\rho \vee W_{\rho'}$ is the class of pairs with $y \leq (\rho\rho')^k x$ for some k, and that it is the class of the least common abstraction iff the alternation $\rho\rho'\rho\rho'\cdots$ is finite at every point; cancellation is needed when meets are not preserved. On presheaves on the path category of a directed graph, a set S of vertices gives the topology whose sheaves are determined at S. For two sets S, T the join of the two inverted classes is the class of the join topology iff the alternations between S and T along paths avoiding $S \cap T$ are bounded; for a finite graph, iff no strongly connected component of the graph minus $S \cap T$ meets both sets, a linear-time test. The join topology is that of $S \cap T$ iff no infinite backward walk avoiding $S \cap T$ alternates between the two sets. When the alternation is bounded by a, exactly $a + 1$ rounds of alternating reconstruction from the values at T and at S give the reconstruction from $S \cap T$.

1. Introduction

In one sentence: for two abstractions of the same concrete lattice, the join of the two classes of "pairs the abstraction cannot tell apart" is generated by alternating the two abstractions and cancelling, and it is the class of the least common abstraction exactly when the alternation is finite at every point; on the path category of a control-flow graph, with abstractions given by sets of cut points, the finiteness is a condition on closed walks, tested in linear time.

Let $\mathcal{C}$ be a category. A *sameness* on $\mathcal{C}$ is a class of morphisms containing the identities and closed under composition and two-out-of-three;

2020 *Mathematics Subject Classification.* 18F10 (primary), 06A15, 68Q55, 18B25 (secondary).

Key words and phrases. closure operator, abstract interpretation, abstract domain, reduced product, Grothendieck topology, presheaf topos, control-flow graph, cut point, two-out-of-three, join of localisations.

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samenesses form a complete lattice $\mathrm{Same}(\mathcal{C})$, the meet being the intersection and the join the least sameness containing both [10]. Every localisation of $\mathcal{C}$ contributes the class W of maps it inverts, and the question studied in [11, 12, 13, 14] is whether W sends the join of two localisations to the join of the two classes. For localisations given by closure operators on subobjects (Lawvere–Tierney topologies, hereditary torsion theories, nuclei) the answer of [14] is a finite-iteration criterion: in a Grothendieck topos or a Grothendieck abelian category, $W_c \vee W_{c'} = W_{\bar{c}}$ for the least closure operator $\bar{c}$ above c and c' iff $(cc')^k = \bar{c}$ for some k.

This paper applies the criterion in two settings suggested by the semantics of programs. The first is the lattice of abstract domains of abstract interpretation [2]: an abstract domain on a complete lattice L of concrete properties is an upper closure operator ρ on L, the closure operators form a complete lattice $\mathrm{uco}(L)$ [16], the reduced product of two domains is their meet and their least common abstraction is their join [5]. The category is the poset L itself, so that the maps are the inequalities $x \leq y$, and the class W_ρ consists of the inequalities that ρ collapses, $\rho x = \rho y$: the refinements y of x that the analysis cannot see. General abstract domains are not nuclei, they need not preserve binary meets (intervals, octagons, reachability under a transition relation), and the criterion of [14] for nuclei on a frame does not apply to them as it stands. Theorem 2.3 treats arbitrary closure operators on a complete lattice: the meet is always preserved, the join is the class of pairs $x \leq y$ with $y \leq (\rho\rho')^k x$ for some k, it is the class of $\rho \sqcup \rho'$ iff the alternation reaches $\rho \sqcup \rho'$ at a finite stage at every x, and, unlike the case of nuclei, the two-out-of-three cancellation is needed to generate the join from the alternating chains (already on the four-element Boolean lattice). Section 3 spells this out for the reachability closures $S \mapsto R^*(S \cup I)$ of transition relations: the join of the two classes is bounded alternation of the two relations, it is the class of the union relation iff the alternation depth is finite from every set of states, which holds for finite state spaces with bound the number of states and fails on $\mathbb{N}$ for the even and the odd steps.

The second setting is the presheaf topos on the path category of a directed graph G, a control-flow graph: a presheaf is a flow of sets along the control flow, and a set S of vertices, a set of cut points in the sense of Bourdoncle [1], determines the Grothendieck topology J_S whose sheaves are the presheaves determined by their values at S, $\mathrm{Sh}(J_S) \simeq \mathrm{Psh}(S)$. Theorem 4.3 says that $W_{J_S} \vee W_{J_T} = W_{J_S \vee J_T}$ iff the number of alternations between visits to T and visits to S along paths that avoid $S \cap T$ is bounded; for a finite graph, iff no closed walk avoiding $S \cap T$ meets both S and T, i.e. no strongly connected component of G with $S \cap T$ removed meets both, a condition decided in linear time; the bound is then $\min(|S \setminus T|, |T \setminus S|)$ and the join topology is $J_{S \cap T}$. Theorem 4.7 identifies the join topology in general: it is $J_{S \cap T}$ iff no infinite backward walk avoiding $S \cap T$ visits both sets infinitely often, a condition weaker than bounded alternation which coincides with it on finite graphs. Theorem 4.10 describes the iterated reconstructions

$(a_S a_T)^k X$ as limits over the paths with k alternations and shows that, when the alternation is bounded by a, the sheafification for the join is $(a_S a_T)^{a+1}$, with $a+1$ sharp: for cut-point topologies the finite-step factorisation of the join sheafification asked for in [14] holds with exactly the number of steps of the closure operators. The theorem interpolates between two known facts: on a finite category every pair of topologies has a preserved join [12], which is the acyclic case, and on the poset $\mathbb{N}$ the topologies of the even and of the odd numbers do not [11], which is the infinite path with unbounded alternation. A third section records the finite action monoids: there the ideal topologies are indexed by sets of idempotent actions, resets, and the reading of the closure is "every future through a reset lands in the invariant".

What is known. Samenesses are from [10]; the finite-iteration criterion in a Grothendieck topos from [13], in exact categories with closure operators, including nuclei on a frame as the poset case, from [14]; the finite-category join theorem from [12]; the counterexample on $\mathbb{N}$ from [11]. Abstract domains as closure operators, the lattice of abstract domains, the reduced product and the least common abstraction are in [2, 16, 5, 6], and the computation of the join of two closure operators by transfinite alternation (Lemma 2.1) is standard and is recalled only to fix notation; the iterated reduction computing the reduced product on a product domain is in [7, 3]; cut sets of control-flow graphs in [1]. Lawvere–Tierney topologies, sheafification and the bidense description of the maps it inverts are in [9, 8].

What is new. Theorem 2.3 (closure operators on a complete lattice without meet preservation: meet, join as alternating chains plus one cancellation, criterion, necessity of cancellation), Proposition 3.1 (reachability closures) and Theorem 4.3 (cut-point topologies on the path category of a graph, with the closed-walk criterion and its linear-time test), Theorem 4.7 (the join of two cut-point topologies is the topology of $S \cap T$ iff no infinite backward walk alternates between the two sets; on a finite graph this is the same condition), Theorem 4.10 (the join sheafification is $a+1$ rounds of alternating reconstruction), together with the dictionary of Section 5. The engine is the criterion of [14]; what is added is its application to closures that are not nuclei and to the object-generated topologies of infinite path categories, where the combinatorics of alternation along paths decides the question.

2. Closure operators on a complete lattice

Let L be a complete lattice, regarded as a category with a morphism $x \to y$ iff $x \leq y$. A *closure operator* on L is a map $\rho\colon L \to L$ that is extensive ($x \leq \rho x$), monotone and idempotent; its set of fixed points $\mathrm{Fix}\rho = \rho(L)$ is closed under arbitrary meets, and ρx is the least fixed point above x. The closure operators on L form a complete lattice $\mathrm{uco}(L)$ under the pointwise order [16], which is the reverse inclusion of fixed-point sets; the meet of ρ, ρ' is $x \mapsto \rho x \wedge \rho' x$, with fixed points the meets $a \wedge b$, $a \in \mathrm{Fix}\rho$, $b \in \mathrm{Fix}\rho'$,

and the join $\rho \sqcup \rho'$ is the closure operator with fixed points $\mathrm{Fix}\rho \cap \mathrm{Fix}\rho'$. In abstract interpretation [2, 5], $\rho \leq \rho'$ means that ρ is more precise, the meet is the reduced product and the join the least common abstraction.

Lemma 2.1 (standard)**.** *For closure operators ρ, ρ' and $x \in L$ define $x_0 = x$, $x_{\alpha+1} = \rho\rho' x_\alpha$ and $x_\lambda = \bigvee_{\alpha<\lambda} x_\alpha$ at limits. The sequence is increasing, stabilises at some ordinal, and its value is $(\rho \sqcup \rho')x$.*

Proof. If $x_{\alpha+1} = x_\alpha =: z$ then $\rho' z \leq \rho\rho' z = z$ gives $\rho' z = z$ and then $\rho z = \rho\rho' z = z$, so $z \in \mathrm{Fix}\rho \cap \mathrm{Fix}\rho'$; every common fixed point above x is above all x_α, so z is the least one. □

Definition 2.2. For a closure operator ρ on L put

$$W_\rho := \{x \leq y : \ \rho x = \rho y\} = \{x \leq y : \ y \leq \rho x\},$$

the inequalities collapsed by ρ, i.e. the refinements y of x that ρ does not distinguish from x.

W_ρ is a sameness on the poset L: it contains the identities, is closed under composition ($x \leq y \leq z$ with $z \leq \rho y \leq \rho\rho x = \rho x$), and under both cancellations ($x \leq y \leq z$ with $z \leq \rho x$ gives $y \leq \rho x$ and $z \leq \rho x \leq \rho y$). No meet preservation is used. If $\rho \leq \rho'$ then $W_\rho \subseteq W_{\rho'}$: a less precise analysis collapses more.

Theorem 2.3. *Let ρ, ρ' be closure operators on a complete lattice L.*

(i) *$W_\rho \cap W_{\rho'} = W_{\rho\sqcap\rho'}$: the reduced product distinguishes $x \leq y$ iff one of the two abstractions does.*

(ii) *$W_\rho \vee W_{\rho'} = \{x \leq y : \ y \leq (\rho\rho')^k x$ for some $k \geq 0\}$ in $\mathrm{Same}(L)$; for such a pair, with $z = (\rho\rho')^k x$, both $x \leq z$ and $y \leq z$ are alternating chains of steps in $W_{\rho'}$ and W_ρ, so every element of the join is obtained from two alternating chains by one cancellation.*

(iii) *$W_\rho \vee W_{\rho'} = W_{\rho\sqcup\rho'}$ iff for every $x \in L$ there is k with $(\rho\rho')^k x = (\rho \sqcup \rho')x$: the alternation of Lemma 2.1 stops at a finite stage at every point.*

(iv) *$\rho\rho' = \rho \sqcup \rho'$ (one alternation suffices uniformly) iff $\rho\rho'$ is idempotent iff ρ maps $\mathrm{Fix}\rho'$ into $\mathrm{Fix}\rho'$.*

(v) *The compositional closure of $W_\rho \cup W_{\rho'}$, the class of alternating chains, can be strictly smaller than the join; it equals the join when ρ and ρ' preserve binary meets.*

Proof. (i) $x \leq y$ lies in $W_{\rho\sqcap\rho'}$ iff $y \leq \rho x \wedge \rho' x$ iff $y \leq \rho x$ and $y \leq \rho' x$.

(ii) Let D be the right-hand side. D is a sameness: if $y \leq (\rho\rho')^k x$ and $z \leq (\rho\rho')^l y$ then $z \leq (\rho\rho')^{k+l} x$; if $x \leq y \leq z$ and $z \leq (\rho\rho')^k x$ then $y \leq (\rho\rho')^k x$ and $z \leq (\rho\rho')^k x \leq (\rho\rho')^k y$. It contains W_ρ and $W_{\rho'}$ ($y \leq \rho x \leq \rho\rho' x$, $y \leq \rho' x \leq \rho\rho' x$). So the join J of W_ρ and $W_{\rho'}$ is contained in D. Conversely let $x \leq y \leq z := (\rho\rho')^k x$, and put $x_0 = x$, $x_{2i+1} = \rho' x_{2i}$, $x_{2i+2} = \rho x_{2i+1}$, so that $x_{2k} = z$. The chain $x = x_0 \leq x_1 \leq \cdots \leq x_{2k} = z$ has its steps alternately in $W_{\rho'}$ and W_ρ, so $(x, z) \in J$. The chain $y = y \vee x_0 \leq y \vee x_1 \leq$

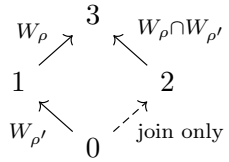
$\cdots \le y \vee x_{2k} = z$ (the last equality as $y \le z$) has the same property: if $x_{j+1} = dx_j$ with $d \in \{\rho, \rho'\}$, then $y \vee x_{j+1} \le d(y \vee x_j)$, both y and dx_j being below $d(y \vee x_j)$; so $(y, z) \in J$. Then $(x, y) \in J$ by cancellation from (x, z) and (y, z).

(iii) $W_{\rho \sqcup \rho'} = \{x \le y : y \le (\rho \sqcup \rho')x\}$ contains D, since $(\rho\rho')^k x \le (\rho \sqcup \rho')x$. If every fibre top $(\rho \sqcup \rho')x$ is some $(\rho\rho')^k x$ the two classes agree; if for some x no $(\rho\rho')^k x$ equals $(\rho \sqcup \rho')x$, then $(x, (\rho \sqcup \rho')x)$ lies in $W_{\rho \sqcup \rho'}$ but not in D, as $(\rho \sqcup \rho')x \le (\rho\rho')^k x$ would force equality.

(iv) If $\rho\rho'$ is idempotent and $a \in \mathrm{Fix}\rho'$, then $\rho'\rho a \le \rho\rho'\rho a = \rho\rho'\rho\rho' a = \rho\rho' a = \rho a$, so $\rho a \in \mathrm{Fix}\rho'$. If ρ preserves $\mathrm{Fix}\rho'$ then $\rho\rho' x \in \mathrm{Fix}\rho'$, so $\rho\rho'\rho\rho' x = \rho\rho\rho\rho' x = \rho\rho' x$. An idempotent $\rho\rho'$ is a closure operator above ρ and ρ' and below every closure operator above both, so it is $\rho \sqcup \rho'$; conversely $\rho \sqcup \rho'$ is idempotent.

(v) Example 2.4 gives a strict inclusion. If ρ, ρ' preserve binary meets, let $x \le y \le z = x_{2k}$ with the chain x_j of (ii) and put $u_j = y \wedge x_j$, so $u_0 = x$, $u_{2k} = y$ and $u_j \le u_{j+1}$. If $x_{j+1} = dx_j$ then $u_{j+1} = y \wedge dx_j \le dy \wedge dx_j = d(y \wedge x_j) = du_j$, so $u_j \le u_{j+1}$ lies in W_d and $x = u_0 \le \cdots \le u_{2k} = y$ is an alternating chain. (When L is a frame this is the poset case of Theorem A of [14].) □

Example 2.4. Let $L = \{0 < 1, 2 < 3\}$ be the four-element Boolean lattice, ρ the closure with $\mathrm{Fix}\rho = \{0, 3\}$ ($\rho 1 = \rho 2 = 3$) and ρ' the closure with $\mathrm{Fix}\rho' = \{1, 3\}$ ($\rho' 0 = 1$, $\rho' 2 = 3$). Then $W_\rho = \{1 \le 3, 2 \le 3\}$, $W_{\rho'} = \{0 \le 1, 2 \le 3\}$ (plus identities), $\rho\rho' 0 = 3$, so $0 \le 2$ lies in the join by (ii); but the only non-identity step out of 0 in $W_\rho \cup W_{\rho'}$ is $0 \le 1$, and $1 \not\le 2$, so $0 \le 2$ is not an alternating chain: it is obtained from $0 \le 3$ and $2 \le 3$ by cancellation. Here $\rho(1 \wedge 2) = 0 \ne 3 = \rho 1 \wedge \rho 2$.



Remark 2.5. In a finite lattice the alternation of Lemma 2.1 stops after at most $|L|$ steps, so (iii) always holds there and the content of the theorem is (ii) and (v): the join is generated by alternating chains *and* cancellation. On an infinite lattice (iii) can fail; the simplest instance is in Section 3.

3. Reachability closures

Let X be a set of states, $R \subseteq X \times X$ a transition relation and $I \subseteq X$ a set of initial states. On the complete lattice $L = \mathcal{P}(X)$ put

$$\rho_{R,I}(S) := R^*(S \cup I),$$

the set of states reachable from S or from I along R: the collecting semantics of "run R from S".

Proposition 3.1. (i) *$\rho_{R,I}$ is a closure operator on $\mathcal{P}(X)$ with fixed points the R-closed sets containing I (the inductive invariants); it preserves binary intersections only in degenerate cases.*

(ii) *$\rho_{R,I} \sqcap \rho_{R',I'}$ is $S \mapsto R^*(S \cup I) \cap R'^*(S \cup I')$ and $\rho_{R,I} \sqcup \rho_{R',I'} = \rho_{R\cup R',I\cup I'}$; for $k \geq 1$, $(\rho_{R,I}\rho_{R',I'})^k S$ is the set of states reachable from $S \cup I'$ by a path of at most k blocks, each an R'-path followed by an R-path, together with the states reachable from I by an R-path followed by at most $k-1$ such blocks (the initial states of the first system enter only after the R'-phase of a block).*

(iii) *$W_{\rho_{R,I}} \vee W_{\rho_{R',I'}}$ is the class of pairs $S \subseteq T$ with $T \subseteq (\rho_{R,I}\rho_{R',I'})^k S$ for some k, i.e. every state of T is reachable in the sense of (ii) with a bounded number of blocks, and it equals $W_{\rho_{R\cup R',I\cup I'}}$ iff the alternation depth of $(R \cup R')$-reachability is finite from every S. If X is finite this holds with at most $|X|$ blocks.*

(iv) *On $X = \mathbb{N}$ with $R = \{(2i, 2i+1)\}$, $R' = \{(2i+1, 2i+2)\}$ and $I = I' = \emptyset$, $(\rho_R\rho_{R'})^k\{0\} = [0, 2k-1]$ for $k \geq 1$ (the first R'-step from 0 is empty), which never reaches $\rho_{R\cup R'}\{0\} = \mathbb{N}$, so $W_{\rho_R} \vee W_{\rho_{R'}} \subsetneq W_{\rho_{R\cup R'}}$.*

Proof. (i) Extensive, monotone and idempotent since R^* is reflexive and transitive; the fixed points are the sets closed under R and containing I. For $x \xrightarrow{R} z \xleftarrow{R} y$ with $x \neq y$ one has $z \in \rho\{x\} \cap \rho\{y\}$ but $\rho(\{x\} \cap \{y\}) = \rho\emptyset = R^*I$. (ii) The meet is pointwise; the common fixed points of the two closures are the sets closed under R and R' containing $I \cup I'$, the fixed points of $\rho_{R\cup R',I\cup I'}$, which is therefore the join. For the iterates, $\rho_{R,I}\rho_{R',I'}(A) = R^*R'^*(A \cup I') \cup R^*I$; writing $B_k(A)$ for the states reachable from A by at most k blocks, so that $B_1B_k = B_{k+1}$ and $B_k(A) \subseteq B_{k+1}(A)$ as blocks may be empty, induction on k gives $(\rho_{R,I}\rho_{R',I'})^k S = B_k(S \cup I') \cup B_{k-1}(R^*I)$. (iii) is Theorem 2.3(ii), (iii); if X is finite, every block that changes the set adds a state, so the iteration of Lemma 2.1 stops after at most $|X|$ blocks. (iv) is a direct computation, and Theorem 2.3(iii). □

Remark 3.2. Example 2.4 is of this form: $X = \{p, q\}$, $R = \{(p, q), (q, p)\}$, $I = \emptyset$ and $R' = \emptyset$, $I' = \{p\}$. The pair $\emptyset \subseteq \{q\}$ is collapsed by the join (from $\emptyset$ the second system reaches p and the first then q) but by no alternating chain of collapses, the only refinement of $\emptyset$ collapsed by one of the two systems being $\{p\} \not\subseteq \{q\}$.

Remark 3.3. The iterated reduction of Granger [7] and of Cousot, Cousot and Mauborgne [3], which alternately refines the two components of a pair (a, b) in a product of abstract domains and is the Nelson–Oppen procedure when the domains are logical theories, computes the reduced product, the *meet* $\rho \sqcap \rho'$, by a decreasing iteration on the product domain. The alternation $\rho\rho'\rho\rho' \cdots$ of Lemma 2.1 computes the *join*, the least common abstraction, by an increasing iteration on L. The two are not to be confused: the first

terminates in one step in the sense of (i) of Theorem 2.3, the second is governed by (iii).

4. Cut-point topologies on the path category of a graph

Let G be a directed graph, finite or infinite, with vertex set V, and let $\mathcal{P}(G)$ be its path category: objects the vertices, morphisms the finite paths, identities the empty paths. A presheaf X on $\mathcal{P}(G)$ is a set $X(c)$ for each vertex and a restriction $X(c) \to X(d)$, $x \mapsto x \cdot f$, for each path $f\colon d \to c$, a flow of sets along the control flow; a subpresheaf is a family of subsets closed under restriction. Write $\mathcal{E} = \mathrm{Psh}(\mathcal{P}(G))$, a Grothendieck topos.

For a set $S \subseteq V$ of vertices, cut points, let D_S be the set of paths that pass through a vertex of S, the empty paths at vertices of S included. D_S is a two-sided ideal of $\mathcal{P}(G)$ (closed under composition with arbitrary paths on either side) and idempotent, $D_S \circ D_S = D_S$, because $f = f \circ \mathrm{id}_s$ for a path f through $s \in S$. For a vertex c let $D_S(c)$ be the sieve of paths into c through S.

Lemma 4.1. *The sieves containing $D_S(c)$, for all c, form a Grothendieck topology J_S on $\mathcal{P}(G)$. Its sheaves are the presheaves X with $X \cong \mathrm{Ran}_i(X|_S)$ for the inclusion i of the full subcategory of $\mathcal{P}(G)$ on the vertices of S, so that restriction gives $\mathrm{Sh}(J_S) \simeq \mathrm{Psh}(S)$, where $\mathrm{Psh}(S)$ denotes the presheaves on that full subcategory (its morphisms are all paths of G between vertices of S, not only the edges inside S), and its closure operator on subpresheaves $A \subseteq X$ is*

$$c_S(A)(c) = \{x \in X(c) :\ x \cdot f \in A \text{ for every path } f\colon s \to c \text{ with } s \in S\}.$$

For two sets S, T one has $c_S c_T = c_{D_S \circ D_T}$, where $c_D(A)(c) = \{x : x \cdot f \in A\ \forall f \in D(c)\}$ for an ideal D; so $(c_S c_T)^k = c_{(D_S \circ D_T)^k}$, and $D_S \circ D_T$ consists of the paths with a visit to T followed by a visit to S.

Proof. Stability: the pullback of $D_S(c)$ along $g\colon e \to c$ is $\{h : gh \in D_S(c)\} \supseteq D_S(e)$. Transitivity: let R be a sieve on c with $h^*R \supseteq D_S(\mathrm{dom}\, h)$ for all $h \in D_S(c)$; for $f = u \circ v \in D_S(c)$ with $u\colon s \to c$, $s \in S$, we have $u \in D_S(c)$ and $\mathrm{id}_s \in D_S(s) \subseteq u^*R$, so $u \in R$ and $f \in R$. Restriction $\mathrm{Psh}(\mathcal{P}(G)) \to \mathrm{Psh}(S)$ is left exact with right adjoint Ran_i, and Ran_i is fully faithful since i is, so restriction is the inverse image of a geometric embedding; a sieve $R \hookrightarrow y(c)$ is inverted by restriction iff R contains every path from a vertex of S into c, i.e. iff $R \supseteq D_S(c)$, so the embedding is $\mathrm{Sh}(J_S)$ [8, A4.3, A4.5]. The closure of A for J_S consists of the x whose sieve $\{f : x \cdot f \in A\}$ is covering, which is the displayed formula; and $c_D c_E(A) = \{x : x \cdot f \in c_E(A)\ \forall f \in D\} = \{x : x \cdot f \cdot g \in A\ \forall f \in D, g \in E\} = c_{D \circ E}(A)$, with $f \circ g$ the path g followed by f. □

Lemma 4.2. *A path f lies in $(D_S \circ D_T)^k$ iff there are vertices $t_1, s_1, \ldots, t_k, s_k$ occurring along f in this order, positions non-decreasing, with $t_i \in T$ and $s_i \in S$. A path through a vertex of $S \cap T$ lies in $(D_S \circ D_T)^k$*

for every k. Hence the chain $(D_S \circ D_T)^k$ stabilises iff the number of alternations "a visit to T, then a visit to S" along paths avoiding $S \cap T$ is bounded, and then $(D_S \circ D_T)^{a+1} = D_{S\cap T}$ for the bound a.

Proof. $f \in (D_S \circ D_T)^k$ iff $f = u_k v_k \cdots u_1 v_1$ with $v_i \in D_T$, $u_i \in D_S$, the splitting points being arbitrary vertices along f and the factors possibly empty; this is the stated condition. At a common vertex x all factors may be taken empty. If the alternation along paths avoiding $S \cap T$ is bounded by a, a path in level $a+1$ passes through $S \cap T$, and conversely such paths are in every level; if it is unbounded, the chain drops at infinitely many levels. □

Theorem 4.3. *Let G be a directed graph and $S, T \subseteq V$. In* $\mathrm{Same}(\mathrm{Psh}(\mathcal{P}(G)))$,

$$W_{J_S} \vee W_{J_T} = W_{J_S \vee J_T}$$

iff the number of alternations between visits to T and visits to S along paths of G avoiding $S \cap T$ is bounded, and then $J_S \vee J_T = J_{S\cap T}$, $\mathrm{Sh}(J_S) \cap \mathrm{Sh}(J_T) \simeq \mathrm{Psh}(S \cap T)$. *If G is finite this holds iff no closed walk of G avoiding $S \cap T$ meets both S and T, i.e. iff no strongly connected component of the induced subgraph $G - (S \cap T)$ contains both a vertex of S and a vertex of T; the bound is then at most* $\min(|S \setminus T|, |T \setminus S|)$, *and the condition is decided in time $O(|V| + |E|)$ by a decomposition into strongly connected components. In particular, for disjoint S, T on a finite graph the join is preserved iff no strongly connected component of G meets both S and T.*

Proof. Let $c = c_S$, $c' = c_T$ and let $\bar{c}$ be the least closure operator above both, the closure operator of the join topology; by [14, Thm. B], $W_{J_S} \vee W_{J_T} = W_{\bar{c}}$ iff $(cc')^k = \bar{c}$ for some k. If the alternation is bounded by a, then by Lemmas 4.1 and 4.2 $(cc')^{a+1} = c_{D_{S\cap T}}$ is idempotent, hence a closure operator above c and c', and every closure operator above both is above cc' and so above $(cc')^{a+1}$; thus $\bar{c} = (cc')^{a+1} = c_{S\cap T}$, the criterion holds, and the join topology is $J_{S\cap T}$. If the alternation is unbounded, the chain of ideals $(D_S \circ D_T)^k$ does not stabilise by Lemma 4.2, and an ideal D is determined by c_D: for $X = y(c)$, $D(c)$ is the least sieve A on c with $\mathrm{id}_c \in c_D(A)(c)$. Since $c, c' \leq \bar{c}$ and $\bar{c}$ is idempotent, $(cc')^k \leq \bar{c}$ for all k and $c\bar{c} = c'\bar{c} = \bar{c}$. If $\bar{c} = (cc')^k$ for some k, then $(cc')^m = (cc')^{m-k}\bar{c} = \bar{c} = (cc')^k$ for all $m \geq k$, so $(D_S \circ D_T)^m = (D_S \circ D_T)^k$ for all $m \geq k$ and the chain stabilises, a contradiction; so the criterion fails. For finite G: if a closed walk avoiding $S \cap T$ meets $s \in S$ and $t \in T$, going round it m times gives alternation $\geq m$; conversely a path avoiding $S \cap T$ with alternation $\geq |S \setminus T| + 1$ repeats an S-vertex, $s_i = s_j$ with $i < j$, with a T-visit t_{i+1} strictly between (a visit to S and one to T at the same position would be a vertex of $S \cap T$), and the subwalk from s_i to s_j is a closed walk avoiding $S \cap T$ meeting S and T; likewise with $T \setminus S$ and the t_i. Two vertices lie on a common closed walk of $G - (S \cap T)$ iff they lie in the same strongly connected component, and the components are computed in linear time [15]. □

Example 4.4. (a) G: $s \to t \to s'$, $S = \{s, s'\}$, $T = \{t\}$. The alternation is at most one, the join is preserved, and $J_S \vee J_T = J_\emptyset$ is the degenerate topology: a J_S-sheaf has $X(t) \cong X(s)$, a J_T-sheaf has $X(s) = 1$ and $X(s') \cong X(t)$, so a common sheaf is terminal. (b) G: the two-cycle $s \rightleftarrows t$, $S = \{s\}$, $T = \{t\}$. Going round the cycle gives unbounded alternation, and $W_{J_S} \vee W_{J_T} \subsetneq W_{J_S \vee J_T}$; adding the vertex s to T restores preservation, as every closed walk then meets $S \cap T = \{s\}$. (b′) G: $s \rightleftarrows u \rightleftarrows t$, $S = \{s\}$, $T = \{t\}$. The simple cycles $s \to u \to s$ and $u \to t \to u$ each meet only one of the two sets, but the closed walk $s \to u \to t \to u \to s$ meets both, the alternation is unbounded and the join is not preserved: the condition of Theorem 4.3 is on closed walks, equivalently on strongly connected components, not on simple cycles. (c) G the infinite path $0 \to 1 \to 2 \to \cdots$, S the even and T the odd vertices: no cycles, unbounded alternation, and the failure of [11].

$$s \longrightarrow t \longrightarrow s' \qquad s \circlearrowright t \qquad s \circlearrowright u \circlearrowright t \qquad 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \cdots$$

Remark 4.5. A finite acyclic G has a finite path category, and [12] gives preservation for all pairs of topologies on a finite category; Theorem 4.3 locates the loops that break it. In the terms of [1], a set of cut points is a set meeting every cycle; the theorem concerns two such sets and asks that their common part meet every closed walk carrying points of both; for one set, meeting every simple cycle and meeting every closed walk are the same condition, for the pair they are not (Example 4.4(b′)). The bound $\min(|S \setminus T|, |T \setminus S|)$ on the alternation is the number of mutual reconstructions, from the values at S and from the values at T, after which the reconstruction from $S \cap T$ is reached.

Remark 4.6. A presheaf on $\mathcal{P}(G)$ restricts against the direction of the edges, $X(c) \to X(d)$ for a path $d \to c$, so the flows of Section 4 are the backward ones; forward flows are presheaves on $\mathcal{P}(G^{\mathrm{op}})$, the graph with the edges reversed. Closed walks and strongly connected components are the same for G and G^{op}, so the criterion of Theorem 4.3 and the bound are the same in both directions.

Theorem 4.7. *Let G be a directed graph and $S, T \subseteq V$. Say that a path* avoids $S \cap T$ *if none of its vertices, endpoints included, lies in $S \cap T$, and call a vertex d* reconstructible *if it lies in the least set $\mathcal{R}$ of vertices such that $d \in \mathcal{R}$ whenever all vertices of S, or all vertices of T, joined to d by a path avoiding $S \cap T$ (the empty path included) lie in $\mathcal{R}$; so the vertices of $S \cap T$ and the vertices with no S-vertex, or no T-vertex, upstream are reconstructible. The following are equivalent:*

(i) $J_S \vee J_T = J_{S \cap T}$, *i.e.* $\mathrm{Sh}(J_S) \cap \mathrm{Sh}(J_T) = \mathrm{Sh}(J_{S \cap T})$;
(ii) *every vertex is reconstructible;*
(iii) *there is no infinite backward walk $v_0 \leftarrow v_1 \leftarrow v_2 \leftarrow \cdots$ in G avoiding $S \cap T$ that visits S and T infinitely often.*

For finite G they hold iff no strongly connected component of $G - (S \cap T)$ meets both S and T, the condition of Theorem 4.3: on a finite graph the

join topology is $J_{S\cap T}$ exactly when the join of the two classes is preserved. In general, bounded alternation implies (i)–(iii), and not conversely.

Proof. Since $D_{S\cap T} \subseteq D_S, D_T$, always $J_S, J_T \leq J_{S\cap T}$, and (i) says that $D_{S\cap T}(d)$ is a covering sieve of $J_S \vee J_T$ for every vertex d.

(ii)⇒(i), by induction along the stages of the least fixed point: suppose that all S-vertices s joined to d by a path avoiding $S \cap T$ already have $D_{S\cap T}(s)$ a join-cover (the case of T is symmetric). $D_S(d)$ is a J_S-cover; for $h \in D_S(d)$ write $h = uv$ with $u\colon s \to d$, $s \in S$. If u meets $S \cap T$ then $h \in D_{S\cap T}(d)$ and $h^*D_{S\cap T}(d)$ is the maximal sieve. Otherwise u avoids $S \cap T$, $D_{S\cap T}(s)$ is a join-cover, and $u^*D_{S\cap T}(d) \supseteq D_{S\cap T}(s)$, so $h^*D_{S\cap T}(d) = v^*u^*D_{S\cap T}(d)$ is a join-cover by stability. By transitivity $D_{S\cap T}(d)$ is a join-cover.

(i)⇒(ii). Let $B = V \setminus \mathcal{R}$. As $\mathcal{R}$ is the least fixed point, every $d \in B$ has a vertex of $S\cap B$ and a vertex of $T\cap B$ joined to d by paths avoiding $S\cap T$. Call a path a *B-path* if it starts in B and avoids $S \cap T$, and let $K(d)$ be the set of sieves R on d such that for every $e \in B$ and every path $f\colon e \to d$ there is a B-path g into e with $fg \in R$. K is a Grothendieck topology: it is upward closed and contains the maximal sieves; it is stable, since for $h\colon d' \to d$ and $f\colon e \to d'$ a B-path g with $hfg \in R$ gives $fg \in h^*R$; it is transitive, since if $Q \in K(d)$ and $h^*R \in K(\operatorname{dom} h)$ for all $h \in Q$, then for $f\colon e \to d$ with $e \in B$ there is a B-path $g_1\colon e_1 \to e$ with $fg_1 \in Q$, and applying $(fg_1)^*R \in K(e_1)$ to the identity of $e_1 \in B$ a B-path g_2 into e_1 with $fg_1g_2 \in R$, where g_1g_2 is a B-path. K contains J_S: for $f\colon e \to d$ with $e \in B$ take $s \in S \cap B$ and a path $g\colon s \to e$ avoiding $S\cap T$; fg passes through s, so $fg \in D_S(d)$. Likewise $K \supseteq J_T$, so $J_S \vee J_T \leq K$. If $d \in B$ then $D_{S\cap T}(d) \notin K(d)$, because for $f = \mathrm{id}_d$ a B-path g avoids $S \cap T$ and is not in $D_{S\cap T}(d)$. So $B \neq \emptyset$ implies $J_{S\cap T} \not\leq K$ and (i) fails.

(ii)⇔(iii). B is the greatest set of vertices outside $S \cap T$ in which every vertex has an S-vertex and a T-vertex of the set upstream along paths avoiding $S \cap T$. From $d \in B$, choosing alternately such a T-vertex and such an S-vertex produces a walk as in (iii); conversely the vertices of such a walk form a set with the property, hence lie in B. For finite G a walk as in (iii) visits some $s \in S$ twice with a visit to T in between, giving a closed walk avoiding $S \cap T$ through S and T, and such a closed walk, traversed repeatedly, is a walk as in (iii); so the condition is that of Theorem 4.3. Finally, bounded alternation gives $(D_S \circ D_T)^{a+1} = D_{S\cap T}$ by Lemma 4.2, so $c_{S\cap T} = (c_Sc_T)^{a+1}$ is the least closure operator above c_S and c_T and the join topology is $J_{S\cap T}$; Example 4.8 shows that the converse fails. □

Example 4.8. (a) For the two-cycle of Example 4.4(b) the comma category of paths from s into t has the edge $s \to t$ as terminal object, so a J_S-sheaf is a presheaf whose restriction along $s \to t$ is a bijection, a J_T-sheaf one whose restriction along $t \to s$ is a bijection, and the common sheaves are the pairs of bijections between two sets, not only the terminal presheaf of $\mathrm{Sh}(J_\emptyset)$: $J_S \vee J_T \neq J_{S\cap T}$, as no vertex is reconstructible. (b) On the infinite path

$0 \to 1 \to 2 \to \cdots$ with S the even and T the odd vertices every vertex is reconstructible, by induction along the path, so $J_S \vee J_T = J_\emptyset$ is degenerate, as computed in [11], although the alternation is unbounded and the join of the two classes is not preserved. (c) On the reversed path $\cdots \to 2 \to 1 \to 0$ with the same S, T no vertex is reconstructible, and the common sheaves are the sequences of bijections $X(0) \cong X(1) \cong X(2) \cong \cdots$.

Remark 4.9. Theorems 4.3 and 4.7 separate two questions that coincide on finite graphs: whether the join of the two topologies is the topology of $S \cap T$, a condition on infinite backward walks, and whether the join of the two classes of inverted maps is the class of the join topology, a uniform bound on finite paths. The first is a co-inductive condition (the greatest set B of the proof), the second an inductive one.

Iterated reconstruction. For a presheaf X the J_S-sheafification $a_S X = \mathrm{Ran}_i(X|_S)$ is the reconstruction of X from its values at S: $(a_S X)(c)$ is the set of compatible families $(x_u)_{u\colon s\to c,\ s\in S}$ with $x_u \in X(s)$ and $x_{ug} = x_u \cdot g$ for every path g between vertices of S, and the unit sends $x \in X(c)$ to $(x \cdot u)_u$. Alternating the reconstructions from T and from S gives the tower $X \to a_S a_T X \to (a_S a_T)^2 X \to \cdots$. The question of [14], whether the sheafification for the join is a finite composite of the two sheafifications and in how many steps, has here a complete answer.

Theorem 4.10. *Let $S, T \subseteq V$ and $k \geq 1$. For a vertex c let $L_k(c)$ be the set of paths $w\colon t \to c$ with $t \in T$ along which there are vertices $t = t_1, s_1, t_2, s_2, \ldots, t_k, s_k$ in this order, positions non-decreasing, with $t_i \in T$ and $s_i \in S$, and let $L'_k(c)$ be the set of those with a further vertex $t_{k+1} \in T$ at or after s_k. Then, naturally in X,*

$$((a_S a_T)^k X)(c) \cong \{(z_w)_{w\in L_k(c)} :\ z_w \in X(t) \text{ for } w\colon t \to c, \\ z_{wg} = z_w \cdot g \text{ for all } g\colon t' \to t \text{ with } t' \in T\},$$

the compatible families of values at T-vertices along the paths with k alternations, and $(a_T(a_S a_T)^k X)(c)$ is the same set with $L'_k(c)$ in place of $L_k(c)$. If the alternation along paths avoiding $S \cap T$ is bounded by a, then $(a_S a_T)^{a+1} X \cong a_{S\cap T} X$, the sheafification for the join topology, naturally in X and compatibly with the units: $a+1$ rounds of alternating reconstruction from T and from S reconstruct from $S \cap T$. If some path avoiding $S \cap T$ has a alternations, as is the case when a is the least bound, then $(a_S a_T)^a$ is not naturally isomorphic to $a_{S\cap T}$: it does not invert the $J_{S\cap T}$-dense monomorphism $D_{S\cap T}(c) \hookrightarrow y(c)$ at the end vertex c of such a path. So, unless $S = T = V$, the least number of rounds after which the alternating reconstruction is the join sheafification is exactly $a+1$, the number of iterations of $c_S c_T$ needed to reach the closure operator of the join (Lemma 4.2).

Proof. Write $\lim_L X$ for the set of families displayed in the statement, for a set L of paths from T-vertices closed under precomposition with paths from T-vertices; it is a presheaf in c by precomposition with paths into c.

With $L'_0(c)$ the set of all paths from T-vertices into c, $\lim_{L'_0} X = a_T X$ by definition. We show $a_S \lim_{L'_k} X \cong \lim_{L_{k+1}} X$; the identity $a_T \lim_{L_k} X \cong \lim_{L'_k} X$ is proved in the same way with S and T exchanged, and the two together give the first two claims by induction on k. Put $Y = \lim_{L'_k} X$. An element of $(a_S Y)(c)$ is a family $(y_u)_{u\colon s\to c,\ s\in S}$ with $y_u = (z^u_w)_{w\in L'_k(s)}$ and $y_{ug} = y_u \cdot g$ for $g\colon s' \to s$, i.e. $z^{ug}_{w'} = z^u_{gw'}$ for $w' \in L'_k(s')$. If a path W into c is split in two ways, $W = uw = u'w'$ with $u\colon s \to c$, $u'\colon s' \to c$ and s' before s along W, then $u' = ug$ and $w = gw'$ for the segment $g\colon s' \to s$, and the condition gives $z^{u'}_{w'} = z^u_w$: the value depends only on the composite path W, and W runs over the paths from T-vertices with a vertex of S at or after t_{k+1}, which are the paths of $L_{k+1}(c)$. The remaining condition $z^u_{wg} = z^u_w \cdot g$, for g from a T-vertex, is the compatibility of the family $(z_W)_{W\in L_{k+1}(c)}$; conversely such a family defines $(y_u)_u$ by $y_u = (z_{uw})_w$. The bijections are natural in X and compatible with the restriction maps and the units.

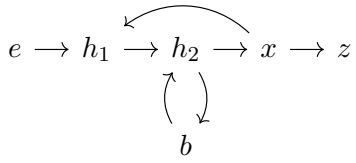
If the alternation along paths avoiding $S \cap T$ is bounded by a, every path of $L_{a+1}(c)$ passes through $S \cap T$, and every path from a vertex of $S \cap T$ lies in $L_{a+1}(c)$ (all t_i, s_i at that vertex). Restriction of families is a map $\lim_{L_{a+1}(c)} X \to (a_{S\cap T} X)(c)$, the latter being the families over the paths from $S \cap T$. It is bijective: given a family (x_v) over the paths v from $S \cap T$ into c, a path $W \in L_{a+1}(c)$ factors as $W = vg$ with v from a vertex $x \in S\cap T$ on W and g into x, and $z_W := x_v \cdot g$ does not depend on the choice of x, since for a later vertex x' of $S \cap T$ on W, with $v = v'h$, $x_v = x_{v'} \cdot h$; this z is compatible and restricts to x, and it is the only such extension. The join topology is $J_{S\cap T}$ (Theorem 4.3), so $a_{S\cap T}$ is its sheafification, and the composite of the units $X \to (a_S a_T)^{a+1} X$, $x \mapsto (x \cdot W)_W$, restricts to the unit $x \mapsto (x \cdot v)_v$ of $a_{S\cap T}$.

For the last claim let $w\colon t \to c$ avoid $S \cap T$ and have a alternations, and let $A = D_{S\cap T}(c) \hookrightarrow y(c)$, the least $J_{S\cap T}$-covering sieve on c, which $a_{S\cap T}$ inverts. If $a = 0$ then $(a_S a_T)^0$ is the identity and $A \neq y(c)$ as $c \notin S \cap T$. If $a \geq 1$, $(a_S a_T)^a A \to (a_S a_T)^a y(c)$ is, by the first part, the inclusion of the families $(z_v)_{v\in L_a(c)}$ with $z_v \in A(t)$ into those with $z_v \in y(c)(t)$; the family $z_v = v$, the image of id_c under the unit, is not in the first set, because $w \in L_a(c)$ avoids $S \cap T$ and so $w \notin A(t)$. Thus $(a_S a_T)^a$ does not invert $A \hookrightarrow y(c)$, and a natural isomorphism $(a_S a_T)^a \cong a_{S\cap T}$ is impossible. If $S = T = V$ both sheafifications are the identity and no round is needed; otherwise some vertex is outside $S \cap T$, the least bound a is attained by some path, and the least number of rounds is $a + 1$. □

Remark 4.11. Theorem 4.10 contrasts with the commuting pairs of [14], where the closure operators compose in one step, $cc' = \bar{c}$, but $a_c a_{c'}$ is not the join sheafification. For cut-point topologies the sheafification tower and the closure tower have the same length. In the language of program analysis: summarising a flow at the cut points T, then at S, and repeating, yields after $a + 1$ rounds the summary at the common cut points, where a is the

maximal number of alternations along the control flow outside $S \cap T$; on a finite graph satisfying the condition of Theorem 4.3, $a \leq \min(|S \setminus T|, |T \setminus S|)$.

Example 4.12. Let G be the control-flow graph of two nested loops: an entry e, the outer header h_1, the inner header h_2, the inner body b, the inner exit x and a final vertex z, with edges $e \to h_1$, $h_1 \to h_2$, $h_2 \to b$, $b \to h_2$, $h_2 \to x$, $x \to h_1$ and $x \to z$.



The set $S = \{h_1, h_2\}$ of loop headers is a cut set in the sense of [1]; let $T = \{h_2, x\}$, so $S \cap T = \{h_2\}$. Every closed walk passes through h_2, so Theorem 4.3 applies: a flow that is determined by its values at the headers and also by its values at $\{h_2, x\}$ is determined by its value at h_2, and the join of the two classes of inverted maps is the class of maps bijective at h_2. The paths avoiding h_2 are $e \to h_1$, $x \to h_1$, $x \to z$ and the empty ones, with at most one alternation ($x \in T$ then $h_1 \in S$), so $a = 1$ and, by Theorem 4.10, two rounds of reconstruction from $\{h_2, x\}$ and then from the headers reconstruct from h_2, while one round does not. With $T = \{x\}$ instead, the outer closed walk $h_1 \to h_2 \to x \to h_1$ meets S and T and avoids $S \cap T = \emptyset$: the join of the two classes is not the class of the join topology, and no finite number of rounds of alternating reconstruction is the join sheafification compatibly with the units, since the pullback of $(a_S a_T)^n A \subseteq (a_S a_T)^n X$ along the unit is $(c_S c_T)^n(A)$ and the operators $(c_S c_T)^n$ never reach the closure operator of the join (proof of Theorem 4.3).

5. Finite action monoids

A deterministic transition system with action alphabet Σ is a coalgebra $X \to X^\Sigma$, i.e. a right Σ^*-set, and the coalgebra homomorphisms are the equivariant maps; the presheaf topos of right M-sets for a monoid M is the general form. Its *ideal topologies* are those with a least covering sieve, and they correspond to the idempotent two-sided ideals $D \subseteq M$, with closure $c_D(A) = \{x : x \cdot D \subseteq A\}$ [13, 14]: a state is in the closure of the invariant A iff all its D-futures lie in A.

Lemma 5.1. *In a finite monoid M a two-sided ideal D is idempotent iff it is generated by idempotents, $D = M\,E\,M$ with $E = \{e \in D : e^2 = e\}$.*

Proof. $(MeM)(MeM) \ni ue\,ev = uev$ gives $MEM \subseteq (MEM)^2 \subseteq MEM$. Conversely let $D = D^2$, so $D = D^n$ for all n, and write $d \in D$ as $d = s_1 \cdots s_n$ with $s_i \in D$ and $n > |M|$. Two prefixes coincide, $s_1 \cdots s_i = s_1 \cdots s_j$ with $i < j$, so with $w = s_{i+1} \cdots s_j \in D$ the prefix $p = s_1 \cdots s_i$ satisfies $p = pw^m$ for all m; some power $e = w^m$ is idempotent, $e \in D$, and $d = pe\,s_{j+1} \cdots s_n \in MeM$. □

Proposition 5.2. *On the right M-sets of a finite monoid M the ideal topologies are indexed by the ideals generated by sets E of idempotents, with $c_{MEM}(A) = \{x : x\,u\,e \in A \text{ for all } u \in M, e \in E\}$: a state is in the closure of the invariant A iff every future that passes through a reset $e \in E$ lands in A. By* [13]*, $W_D \vee W_{D'} = W_{J_D \vee J_{D'}}$ for any two of them, with the stabilisation rank of $(DD')^k$ as the bound on the alternation. For the free monoid Σ^*, and more generally for a monoid with a length function additive and positive on non-identity elements, the only idempotent ideals are $\emptyset$ and M, so the deterministic systems over a free alphabet have no non-trivial ideal topology.*

Proof. The first statement is Lemma 5.1 and the description of c_D, as $x \cdot MeM \subseteq A$ iff $x\,u\,e \in A$ for all u (the right factor preserves A). The second is [13] together with finiteness. For the last, a shortest non-identity element of an idempotent ideal $D \neq M$ is a product of two elements of D of larger length. □

Remark 5.3. Nontrivial ideal topologies on deterministic systems thus need relations among the actions, typically idempotent actions (resets, normalisation steps). In a band ($w^2 = w$ for all w) every ideal is idempotent, and for the free band on Σ the ideals D_a of words containing the letter a satisfy $D_a D_b = D_a \cap D_b$: one alternation. The contracted cycle monoids of [12], with stabilisation rank growing with the cycle, are the instances of unbounded alternation in this setting.

6. Questions

Question 6.1. Which pairs of standard numerical abstract domains (intervals, congruences, octagons, polyhedra, on $\mathcal{P}(\mathbb{Z}^n)$) have unbounded alternation at some element, so that the join of the two classes of collapsed pairs is not the class of the least common abstraction? The pairs checked (intervals and congruences) stabilise in two steps because the interval hull of an infinite congruence class is a product of lines and points.

Question 6.2. Extend Theorem 4.3 to flows with transfer functions, i.e. to presheaves on a quotient of the path category by relations among the edges (idempotent edges, commuting edges), where the ideals D_S are replaced by their images; for which relations does the cycle criterion survive?

Funding

No funding was received for this work.

Declarations

The author has no competing interests to declare. The finite checks mentioned in the text (closure operators on small lattices, random transition systems, the component criterion on random graphs, the number of rounds of iterated reconstruction on random acyclic graphs) are the files `uco_same.py`, `reach_same.py`, `cutpoints.py` and `iter_recon.py` in

the directory `scripts/comparison` of the repository `https://github.com/r-shibusawa/cubical-strict-j`, archived on Zenodo under the concept DOI 10.5281/zenodo.21405961.

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